

Assignment #1

Big Data Processing Using Apache Spark

Course/Code: **Big Data System Design (MBAI 5110G)**

Semester/Year: **Winter-2026**

Professor/Instructor: **Dilli Prasad Sharma, PhD**

Submission Due: **15 February 2026 (End of day)**

Learning Objectives

After completing this assignment, students will be able to:

- Load and process large-scale datasets using Spark Data Frames
- Write analytical queries using Spark SQL
- Implement real-time processing using Structured Streaming
- Perform low-level transformations using RDDs
- Compare different Spark abstractions in terms of performance and usability

Submission Instructions

All students **MUST** submit this assignment individually via Canvas, uploading a **single ZIP file** containing the **report** and **source code**. Follow the instructions below carefully:

1. Report

- Prepare a **short and effective report** including:
 - **Cover page** with the information: Course Name, Course Code, Term & Year, Assignment Number, Submission Due Date, Submitted Date, Student Name, Student ID
 - **Body of the report**: Type each question and provide the **corresponding answer**, including Code segments and corresponding output images/screenshots of results
- Save the report as a **single PDF or Word file** with the naming: *<LastName_StudentID_Assignment_1_Report.pdf>*

2. Source Code

- Include your **Jupyter Notebook** or **Python source code** file(s) with all implemented tasks (files: .ipynb or .py). File naming: *<LastName_StudentID_Assignment_1_Code.ipynb>*

3. ZIP File

- Keep **exactly two files** (Report + Source Code).
- Compress them using **ZIP format only** (do not use RAR, 7z, or other compression formats).
- Submit the **single ZIP file** to the **Canvas's Assignment submission folder**. Please follow the file naming: *<LastName_StudentID_Assignment_1.zip>*
Example: Sharma_123456_Assignment_1.zip)

Dataset: NYC Yellow Taxi Trip Records

This dataset contains millions of taxi trips in New York City, including pickup/drop-off times, locations, passenger counts, and fares.

Source: NYC Taxi & Limousine Commission (TLC)

URL: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

Students may download **1 month of data from 2025 to keep computations** manageable. Please specify which month of data you use in your work.

Use the above dataset and answer the following questions.

Part A: Spark DataFrames

(25 Marks)

Task A1: Load and Inspect Data

(5 marks)

1. Load the CSV/Parquet data into Spark DataFrame
2. Print schema and show sample 5 rows
3. Handle missing/null or invalid values and check missing/null values

Task A2: DataFrame Analytics

(20 marks)

1. Print the average trip distance per day
 2. Print the top 10 pickup locations by number of trips
 3. Find the average fare per passenger count
 4. Filter trips longer than 10 miles and analyze fare distribution
-

Part B: Spark SQL

(25 Marks)

Create a Temporary View named '**taxi_trips**', and write SQL queries using it to answer the following questions:

1. Find peak pickup hours
 2. Calculate total revenue per day
 3. Identify the top 5 busiest pickup-drop-off pairs
 4. Detect suspicious trips (distance < 1 mile but fare > \$50)
-

Part C: Spark Structured Streaming

(25 Marks)

1. Create streaming Taxi Trip Dataset: Use the rate source to simulate streaming taxi trip data.
[10 marks]
2. Compute the rolling average fare every 5 minutes.
[5 marks]
3. Count trips per pickup location using window operations.
[5 marks]
4. Write results to:
 - Console

- Memory table
- File (Parquet/JSON)

[5 marks]

Part D: RDD-Operations

(15 Marks)

Perform the following RDD Operations:

- Map: Extract (pickup_location, fare)
- ReduceByKey: Compute the total fare per pickup location
- Filter: Select trips with passenger_count >

Part E: Discussion/Reflection Report

(10 Marks)

Prepare a **1-page discussion/reflection report** about this assignment, addressing the following questions:

1. RDDs vs DataFrames:

Under what circumstances should RDDs be preferred over DataFrames? Justify with examples.

2. Performance Comparison:

Compare the performance characteristics of RDDs, DataFrames, and Spark SQL. Briefly explain how Spark's optimization mechanisms affect performance.

3. Streaming Challenges:

Discuss the key challenges associated with real-time streaming data processing in Spark, such as state management, late data, and fault tolerance.

4. Applications for Structured Streaming:

Describe at least three real-world applications where Structured Streaming is suitable, explaining why it is preferred over batch processing in these cases